

# FGFR2 fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes FGFR2 gene fusions from the msk\_impact\_50k\_2026 study, covering 136 fusion events. 108/136 fusions (79.4%) were in-frame. 115/136 fusions (84.6%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ( $p=0.000424687$ , statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for FGFR2.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 136 analyzed fusion events, identifying 78 distinct fusion partner genes. The most frequent partner was BICC1, observed in 28 events.

### Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 99/108 (91.7%) of in-frame fusions retained the domain,  $p=0.000424687$  (statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

### Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 9/108 (8.3%) of in-frame fusions disrupted the domain,  $p=0.989695$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.792079.

### Cutpoint detection

Cutpoint detection scanned 134 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 767 aa (permutation-corrected  $p=0.00990099$ , statistically significant at  $\alpha=0.05$ ). This is 10 aa from the nearest configured domain boundary at 757 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=115$ ) and "not\_retained" ( $n=19$ ). For Frame\_status, retained: 99/102 (97.1%, 95% CI 91.7%-99.0%); not\_retained: 9/14 (64.3%, 95% CI 38.8%-83.7%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 104.965 vs 61.8947,  $p=0.20677$  (not statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score produced results for this run (78 row(s) in composite\_evidence\_ranking); see the accompanying table.

## Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## Joint partner

Joint partner reported: FGFR2 has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

### composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| BICC1        | 28          | 0.20588235294117646  | 0.20043213737826426    | 0.010123138679069898    | 0.9968060394889663       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.3373289449752655  | 1    |
| TACC2        | 8           | 0.058823529411764705 | 0.20043213737826426    | 0.010123138679069898    | 0.9928861788617886       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.29262331882236536 | 2    |
| KIAA1217     | 4           | 0.029411764705882353 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28486686258133237 | 3    |
| NOL4         | 4           | 0.029411764705882353 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28486686258133237 | 4    |
| SHTN1        | 3           | 0.022058823529411766 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28266098022839115 | 5    |
| TBC1D4       | 3           | 0.022058823529411766 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28266098022839115 | 6    |
| CTNNA3       | 3           | 0.022058823529411766 | 0.20043213737826426    | 0.010123138679069898    | 0.9994579945799458       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28257967941538303 | 7    |
| BAIAP2L1     | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28045509787545    | 8    |
| NRBF2        | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28045509787545    | 9    |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score    | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|--------------------|------|
| OPTN         | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28045509787545   | 10   |
| SORBS1       | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28045509787545   | 11   |
| TTC28        | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28045509787545   | 12   |
| WAC          | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28045509787545   | 13   |
| AHCYL1       | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 0.9991869918699187       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2803331466559378 | 14   |
| INA          | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 0.9983739837398374       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2802111954364256 | 15   |
| ABLIM1       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 16   |
| ADAM9        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 17   |
| AMOT         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 18   |
| ATAD1        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 19   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score    | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|--------------------|------|
| ATP6V1D      | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 20   |
| BICD1        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 21   |
| BNIP2        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 22   |
| BTBD16       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 23   |
| CAT          | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 24   |
| CCDC186      | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 25   |
| CCDC6        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 26   |
| CCDC73       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 27   |
| CEACAM7      | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 28   |
| DDX21        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 29   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score    | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|--------------------|------|
| DIABLO       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 30   |
| GAB2         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 31   |
| HSPA12A      | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 32   |
| KIF14        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 33   |
| L3MBTL3      | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 34   |
| LAMC1        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 35   |
| MARVELD3     | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 36   |
| MBIP         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 37   |
| MYH15        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 38   |
| MYH9         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 39   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score    | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|--------------------|------|
| MYLK         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 40   |
| MYO18B       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 41   |
| NRAP         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 42   |
| PAH          | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 43   |
| PALLD        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 44   |
| PDZRN4       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 45   |
| PKP4         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 46   |
| POC1B        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 47   |
| PPP1R21      | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 48   |
| PRDX3        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 49   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score    | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|--------------------|------|
| RABGAP1L     | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 50   |
| ROCK1        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 51   |
| SEPTIN10     | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 52   |
| SH3GLB1      | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 53   |
| SHC2         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 54   |
| SKI          | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 55   |
| SRRT         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 56   |
| STAT4        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 57   |
| TFEC         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 58   |
| UBP1         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 59   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score    | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|--------------------|------|
| VCL          | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 60   |
| ZFAT         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 61   |
| ZMYM4        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 1.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2782492155225088 | 62   |
| CEP112       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.9983739837398374       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2780053130834844 | 63   |
| KLHL29       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.9983739837398374       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2780053130834844 | 64   |
| NID2         | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.9983739837398374       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2780053130834844 | 65   |
| RNF145       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.9983739837398374       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2780053130834844 | 66   |
| RPAP3        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.9983739837398374       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2780053130834844 | 67   |
| VPS26A       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.9983739837398374       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2780053130834844 | 68   |
| YTHDF3       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.9983739837398374       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2780053130834844 | 69   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| FOXP1        | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 0.9723577235772358       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.27630875641203534 | 70   |
| NRG3         | 2           | 0.014705882352941176 | 0.20043213737826426    | 0.010123138679069898    | 0.9707317073170731       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.27606485397301095 | 71   |
| KCNH7        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.9495934959349593       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2706882399127527  | 72   |
| ANKRD11      | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.943089430894309        | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.26971263015665514 | 73   |
| ATE1         | 3           | 0.022058823529411766 | 0.20043213737826426    | 0.010123138679069898    | 0.8384823848238483       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2584333379519684  | 74   |
| ADAM12       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.8520325203252033       | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25605409357128933 | 75   |
| PRKAG2       | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | None                     | 0.739106710891876          | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.15088143002648097 | 76   |
| ST8SIA6      | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | None                     | 0.739106710891876          | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.15088143002648097 | 77   |
| GRID1        | 1           | 0.007352941176470588 | 0.20043213737826426    | 0.010123138679069898    | 0.0                      | 0.739106710891876          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.1282492155225088  | 78   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio | p_value            | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|------------|--------------------|--------------------|
| 152      | 0                      | 115              | 1                      | 18               | 0.0        | 0.1417910447761194 | 0.8483511974119787 |

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio | p_value                | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|------------|------------------------|--------------------|
| 469      | 0                      | 115              | 2                      | 17               | 0.0        | 0.019189765458422176   | 1.7169303332757584 |
| 676      | 0                      | 115              | 3                      | 16               | 0.0        | 0.0024714091878270984  | 2.607055343103334  |
| 732      | 0                      | 115              | 5                      | 14               | 0.0        | 3.4829019675778254e-05 | 4.458058749354329  |
| 767      | 111                    | 4                | 5                      | 14               | 77.7       | 8.867691233331173e-12  | 11.052189437117653 |
| 768      | 111                    | 4                | 19                     | 0                | 0.0        | 1.0                    | -0.0               |
| 798      | 112                    | 3                | 19                     | 0                | 0.0        | 1.0                    | -0.0               |
| 801      | 113                    | 2                | 19                     | 0                | 0.0        | 1.0                    | -0.0               |
| 802      | 114                    | 1                | 19                     | 0                | 0.0        | 1.0                    | -0.0               |

## domain\_disruption tables

*frame\_domain\_contingency\_table*

|    |    |
|----|----|
| 9  | 6  |
| 99 | 20 |

*domain\_disruption\_descriptives*

| Domain_key              | Domain_name                 | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-------------------------|-----------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_disruption_pf13895 | Immunoglobulin domain       | 134                     | 119                  | 0               | 15               | 0.0                                             |
| auto_disruption_pf07679 | Immunoglobulin I-set domain | 134                     | 118                  | 0               | 16               | 0.0                                             |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|    |    |
|----|----|
| 99 | 16 |
| 9  | 10 |

*domain\_retention\_descriptives*

| Domain_key | Domain_name                                  | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|------------|----------------------------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| kinase     | Protein tyrosine and serine/threonine kinase | 134                     | 115                  | 3               | 16               | 0.09006270188105643                             |

## exon\_retention tables

### Exon\_retention

| Exon | Retained_event_count | Total_event_count | Retained_fraction  |
|------|----------------------|-------------------|--------------------|
| 11   | 117                  | 136               | 0.8602941176470589 |

## frequency tables

### Partner\_gene\_counts

| Partner_gene | Event_count |
|--------------|-------------|
| ABLM1        | 1           |
| ADAM12       | 1           |
| ADAM9        | 1           |
| AHCYL1       | 2           |
| AMOT         | 1           |
| ANKRD11      | 1           |
| ATAD1        | 1           |
| ATE1         | 3           |
| ATP6V1D      | 1           |
| BAIAP2L1     | 2           |
| BICC1        | 28          |
| BICD1        | 1           |
| BNIP2        | 1           |
| BTBD16       | 1           |
| CAT          | 1           |
| CCDC186      | 1           |
| CCDC6        | 1           |
| CCDC73       | 1           |
| CEACAM7      | 1           |
| CEP112       | 1           |
| CTNNA3       | 3           |
| DDX21        | 1           |
| DIABLO       | 1           |
| FOXP1        | 2           |
| GAB2         | 1           |
| GRID1        | 1           |
| HSPA12A      | 1           |
| INA          | 2           |

| Partner_gene | Event_count |
|--------------|-------------|
| KCNH7        | 1           |
| KIAA1217     | 4           |
| KIF14        | 1           |
| KLHL29       | 1           |
| L3MBTL3      | 1           |
| LAMC1        | 1           |
| MARVELD3     | 1           |
| MBIP         | 1           |
| MYH15        | 1           |
| MYH9         | 1           |
| MYLK         | 1           |
| MYO18B       | 1           |
| NID2         | 1           |
| NOL4         | 4           |
| NRAP         | 1           |
| NRBF2        | 2           |
| NRG3         | 2           |
| OPTN         | 2           |
| PAH          | 1           |
| PALLD        | 1           |
| PDZRN4       | 1           |
| PKP4         | 1           |
| POC1B        | 1           |
| PPP1R21      | 1           |
| PRDX3        | 1           |
| PRKAG2       | 1           |
| RABGAP1L     | 1           |
| RNF145       | 1           |
| ROCK1        | 1           |
| RPAP3        | 1           |
| SEPTIN10     | 1           |
| SH3GLB1      | 1           |
| SHC2         | 1           |
| SHTN1        | 3           |
| SKI          | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| SORBS1       | 2           |
| SRRT         | 1           |
| ST8SIA6      | 1           |
| STAT4        | 1           |
| TACC2        | 8           |
| TBC1D4       | 3           |
| TFEC         | 1           |
| TTC28        | 2           |
| UBP1         | 1           |
| VCL          | 1           |
| VPS26A       | 1           |
| WAC          | 2           |
| YTHDF3       | 1           |
| ZFAT         | 1           |
| ZMYM4        | 1           |

### Per-event results (results.tsv)

| event_id                | sample_id         | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                   | lost_domains | disrupted_domains                            | source_annotation_text                                                                                                                                 | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------------|--------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0021826-T01-IM6-1 | P-0021826-T01-IM6 |            | ANKRD11-FGFR2 | ANKRD11      | unknown      | five_prime  | 123243799          | 16              | 732                         | True                   | disrupted     | 0.9097472924187726       | True                | Immunoglobulin domain; Immunoglobulin I-set domain |              | Protein tyrosine and serine/threonine kinase | FGFR2 (NM_000141) - ANKRD11 (NM_013275) rearrangement: t(10;16)(q26.13;q24.3)(chr10:g.123243799::chr16:g.89538175)   Transcript Fusion {FGFR2:ANKRD11} | NA                           |

| event_id                        | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name       | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                                                                                                                                                                              | source_site<br>2_effect_on<br>_frame |
|---------------------------------|---------------------------|----------------------------|-----------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-001<br>3241-T01-I<br>M5-2 | P-0013<br>241-T0<br>1-IM5 |                            | ATP6V<br>1D-FG<br>FR2 | ATP<br>6V1<br>D          | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>048              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>ATP6V1D (NM_015994)<br>rearrangement: t(10;14)(q26.<br>13;q23.2)(chr10:g.12324204<br>8::chr14:g.67818116)  <br>Protein fusion: in frame<br>(FGFR2-ATP6V1D)                                                                                                                                                   | NA                                   |
| EVT-P-000<br>9512-T01-I<br>M5-3 | P-0009<br>512-T0<br>1-IM5 |                            | BICC1-<br>FGFR2       | BIC<br>C1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>449              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | Note: The FGFR2<br>(NM_000141) - BICC1<br>(NM_001080512) reciprocal<br>rearrangement event is an<br>inversion which results in the<br>fusion of FGFR2 exons 1-17<br>with BICC1 exons 3-21 . The<br>fusion is predicted to be in<br>frame and includes the<br>kinase domain of FGFR2.  <br>Protein fusion: in frame<br>(FGFR2-BICC1) | NA                                   |
| EVT-P-000<br>2675-T01-I<br>M3-4 | P-0002<br>675-T0<br>1-IM3 |                            | BICC1-<br>FGFR2       | BIC<br>C1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>096              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | NA   Protein fusion: in frame<br>(FGFR2-BICC1)                                                                                                                                                                                                                                                                                      | NA                                   |

| event_id                    | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_introductory_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                        | source_site2_effect_on_frame |
|-----------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|----------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>9067-T01-IM6-5 | P-0029<br>067-T01-IM6 |            | BICC1-FGFR2 | BICC1        | in-frame     | five_prime  | 123240197          | 17              | 767                         | True                       | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 3-21) : c.2302-67:FGFR2_c.238-6757:BICC1inv   Protein Fusion: in frame {FGFR2:BICC1} | NA                           |
| EVT-P-000<br>3177-T01-IM5-6 | P-0003<br>177-T01-IM5 |            | BICC1-FGFR2 | BICC1        | in-frame     | five_prime  | 123242188          | 17              | 767                         | True                       | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | NA   Protein fusion: in frame (FGFR2-BICC1)                                                                                                                                   | NA                           |
| EVT-P-002<br>5807-T01-IM6-7 | P-0025<br>807-T01-IM6 |            | BICC1-FGFR2 | BICC1        | in-frame     | five_prime  | 123242181          | 17              | 767                         | True                       | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 3-21): c.2301+895:FGFR2_c.238-35107:BICC1inv   Protein Fusion: in frame {FGFR2:BICC1}  | NA                           |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status | tar_get_role | breakp_o_int_ge_nomic | break_p_o_int_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains                                                                                                                                        | lost_dom_ains                                                                                                                                                                   | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                                    | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|--------------|-----------------|---------------|---------------|--------------|-----------------------|---------------------|------------------------------|--------------------------|-----------------|----------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>5250-T01-I<br>M6-8  | P-0025<br>250-T0<br>1-IM6 |              | BICC1-<br>FGFR2 | BIC<br>C1     | in-fr<br>ame  | thr_ee_prime | 123240<br>491         | 18                  | 768                          | True                     | lost            | 0.0                        | False                |                                                                                                                                                           | Prote<br>in tyr<br>osine<br>and<br>serin<br>e/thr<br>eoni<br>ne ki<br>nase<br>; Im<br>mun<br>oglo<br>bulin<br>dom<br>ain; I<br>mmu<br>nogl<br>obuli<br>n<br>I-set<br>dom<br>ain |                     | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>fusion (FGFR2 exons 1-17<br>fused to BICC1 exons<br>16-21): c.2302-956:FGFR2_<br>c.2182-297:BICC1inv  <br>Protein Fusion: in frame<br>{BICC1:FGFR2}        | NA                            |
| EVT-P-003<br>2033-T01-I<br>M6-9  | P-0032<br>033-T0<br>1-IM6 |              | BICC1-<br>FGFR2 | BIC<br>C1     | in-fr<br>ame  | fiv_e_prime  | 123241<br>685         | 17                  | 767                          | True                     | retai<br>ned    | 1.0                        | False                | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                                 |                     | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>Fusion (FGFR2 exon 18<br>fused with BICC1 exon3) : c.<br>2301+1527:FGFR2_c.237+3<br>8877:BICC1inv   Protein<br>Fusion: in frame<br>{FGFR2:BICC1}           | NA                            |
| EVT-P-000<br>5538-T01-I<br>M5-10 | P-0005<br>538-T0<br>1-IM5 |              | BICC1-<br>FGFR2 | BIC<br>C1     | in-fr<br>ame  | fiv_e_prime  | 123242<br>753         | 17                  | 767                          | True                     | retai<br>ned    | 1.0                        | False                | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                                 |                     | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>reciprocal fusion (FGFR2<br>exons 1-17 with BICC1<br>exons 16-21): c.2301+459:F<br>GFR2_c.2182-276:BICC1inv<br>  Protein fusion: in frame<br>(FGFR2-BICC1) | NA                            |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                   | source_site_2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0020186-T01-IM6-11 | P-0020186-T01-IM6 |            | BICC1-FGFR2 | BICC1        | in-frame     | five_prime  | 123241148          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exon17 fused with BICC1 exon3 ): c.2302-1613:FGFR2_c.237+32254: BICC1inv   Protein Fusion: in frame {FGFR2:BICC1}                 | NA                            |
| EVT-P-0020025-T01-IM6-12 | P-0020025-T01-IM6 |            | BICC1-FGFR2 | BICC1        | in-frame     | five_prime  | 123241984          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 with BICC1 exons 3-21) : c.2301+1228:FGFR2_c.238-4773:BICC1inv   Protein Fusion: in frame {FGFR2:BICC1}                | NA                            |
| EVT-P-0018115-T01-IM6-13 | P-0018115-T01-IM6 |            | BICC1-FGFR2 | BICC1        | in-frame     | five_prime  | 123241970          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused in-frame with BICC1 exons 3-21): c.2301+1242:FGFR2_c.237+34947:BICC1dup   Protein Fusion: in frame {FGFR2:BICC1} | NA                            |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                   | lost_domains                                                                                       | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0015942-T01-IM6-14 | P-0015942-T01-IM6 |            | BICC1-FGFR2 | BICC1        | in-frame     | five_prime  | 123241020          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain |                                                                                                    |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exons 1-17 with BICC1 exons 9-21) : c.2302-1485:FGFR2_c.1048-60:BICC1inv   Protein Fusion: in frame {FGFR2:BICC1}        | NA                           |
| EVT-P-0008146-T01-IM5-15 | P-0008146-T01-IM5 |            | BICC1-FGFR2 | BICC1        | in-frame     | five_prime  | 123240154          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain |                                                                                                    |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2302-619:FGFR2_c.2182-744:BICC1inv   Protein fusion: in frame (FGFR2-BICC1) | NA                           |
| EVT-P-0037617-T01-IM6-16 | P-0037617-T01-IM6 |            | BICC1-FGFR2 | BICC1        | in-frame     | three_prime | 123241302          | 18              | 768                         | True                   | lost          | 0.0                      | False               |                                                                                                    | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain |                   | BICC1 (NM_001080512) - FGFR2 (NM_000141) rearrangement: c.238-34060:BICC1_c.2302-1767:FGFR2inv   Protein Fusion: in frame {BICC1:FGFR2}                                         | NA                           |

| event_id                         | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name       | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                                       | source_site<br>2_effect_on<br>_frame |
|----------------------------------|---------------------------|----------------------------|-----------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-001<br>5432-T01-I<br>M6-17 | P-0015<br>432-T0<br>1-IM6 |                            | CAT-F<br>GFR2         | CAT                      | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>534              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) - CAT<br>(NM_001752)<br>rearrangement: t(10;11)(q26.<br>13;p13)(chr10:g.123242534::<br>chr11:g.34467980)   Protein<br>fusion: in frame<br>(FGFR2-CAT)                      | NA                                   |
| EVT-P-002<br>7305-T01-I<br>M6-18 | P-0027<br>305-T0<br>1-IM6 |                            | CCDC<br>6-FGF<br>R2   | CC<br>DC6                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>542              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>CCDC6 (NM_005436) fusion<br>(FGFR2 exons 1-17 fused to<br>CCDC6 exons 4-9): c.2301+<br>1670:FGFR2_c.582+6426:C<br>CDC6del   Protein Fusion: in<br>frame {FGFR2:CCDC6} | NA                                   |
| EVT-P-002<br>7741-T01-I<br>M6-19 | P-0027<br>741-T0<br>1-IM6 |                            | CEAC<br>AM7-F<br>GFR2 | CEA<br>CA<br>M7          | unk<br>now<br>n              | fiv<br>e_<br>pri<br>me  | 123241<br>990              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141)<br>rearrangement: t(10;19)(q26.<br>13;q13.2)(chr10:g.12324199<br>0::chr19:g.42175729)  <br>Transcript Fusion<br>{FGFR2:CEACAM7}                                            | NA                                   |

| event_id                         | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name      | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains                                                                                                                                                           | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                   | source_site<br>2_effect_on<br>_frame |
|----------------------------------|---------------------------|----------------------------|----------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-000<br>3724-T01-I<br>M5-20 | P-0003<br>724-T0<br>1-IM5 |                            | CEP11<br>2-FGF<br>R2 | CEP<br>112               | in-fr<br>ame                 | thre<br>e_<br>pr<br>ime | 123241<br>923              | 18                          | 768                                 | True                           | lost                      | 0.0                             | False                       |                                                                                                                                                           | Prote<br>in tyr<br>osine<br>and<br>serin<br>e/thr<br>eonine<br>ki<br>nase<br>; Im<br>mun<br>oglo<br>bulin<br>dom<br>ain; I<br>mmu<br>nogl<br>obuli<br>n<br>I-set<br>dom<br>ain |                           | t(17;10)(q24.1;q26.13)(chr17:g.64023428::chr10:g.123241923)   Protein fusion: in frame (CCDC46-FGFR2)                                    | NA                                   |
| EVT-P-001<br>9723-T01-I<br>M6-22 | P-0019<br>723-T0<br>1-IM6 |                            | CTNN<br>A3-FG<br>FR2 | CTN<br>NA3               | in-fr<br>ame                 | fiv<br>e_<br>pr<br>ime  | 123243<br>018              | 17                          | 767                                 | True                           | retain<br>ed              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                                |                           | FGFR2 (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194: FGFR2_c.1885-25982:CTNNA3del   Protein Fusion: in frame {FGFR2:CTNNA3} | NA                                   |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusion_name          | part_ner_gene | fra_me_status | tar_get_role | breakp_o_int_ge_nomic | break_p_o_int_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains                                                                                                                                     | lost_dom_ains                                                                                                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                      | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|--------------|----------------------|---------------|---------------|--------------|-----------------------|---------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>1516-T01-I<br>M5-23 | P-0011<br>516-T0<br>1-IM5 |              | CTNN<br>A3-FG<br>FR2 | CTN<br>NA3    | in-fr<br>ame  | thr_ee_prime | 123241<br>940         | 18                  | 768                          | True                     | lost           | 0.0                        | False                |                                                                                                                                                        | Prote<br>in tyr<br>osine<br>and<br>serin<br>e/thr<br>eoni<br>ne ki<br>nase<br>; Im<br>mun<br>oglo<br>bulin<br>dom<br>ain; I<br>mmu<br>nogl<br>obuli<br>n I-set<br>dom<br>ain |                     | CTNNA3 (NM_013266)<br>-FGFR2 (NM_000141)<br>Rearrangement : c.1048-847<br>9:CTNNA3_c.2301+1272:FG<br>FR2dup   Protein fusion: in<br>frame (CTNNA3-FGFR2)                                    | NA                            |
| EVT-P-002<br>0410-T01-I<br>M6-24 | P-0020<br>410-T0<br>1-IM6 |              | DDX21<br>-FGFR<br>2  | DD<br>X21     | in-fr<br>ame  | fiv_e_prime  | 123243<br>048         | 17                  | 767                          | True                     | retai<br>ned   | 1.0                        | False                | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin I-set<br>d omain |                                                                                                                                                                              |                     | FGFR2 (NM_000141) -<br>DDX21 (NM_004728) fusion<br>(FGFR2 exons 1-17 fused to<br>DDX21 exons 2-15 ): c.2301<br>+164:FGFR2_c.88-860:DDX<br>21inv   Protein Fusion: in<br>frame {FGFR2:DDX21} | NA                            |
| EVT-P-001<br>8869-T03-I<br>M6-26 | P-0018<br>869-T0<br>3-IM6 |              | FGFR2<br>-ABLI<br>M1 | ABL<br>IM1    | in-fr<br>ame  | fiv_e_prime  | 123242<br>601         | 17                  | 767                          | True                     | retai<br>ned   | 1.0                        | False                | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin I-set<br>d omain |                                                                                                                                                                              |                     | FGFR2 (NM_000141) -<br>ABLIM1 (NM_002313)<br>fusion: c.2301+611:FGFR2_<br>c.895-12244:ABLIM1del  <br>Protein Fusion: in frame<br>{FGFR2:ABLIM1}                                             | NA                            |

| event_id                         | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name      | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains                                      | source_annotation_text                                                                                                                                                                                                                 | source_site<br>2_effect_on<br>_frame |
|----------------------------------|---------------------------|----------------------------|----------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-002<br>6275-T01-I<br>M6-27 | P-0026<br>275-T0<br>1-IM6 |                            | FGFR2<br>-ADAM<br>12 | ADA<br>M12               | unk<br>now<br>n              | fiv<br>e_<br>pri<br>me  | 123246<br>899              | 15                          | 676                                 | False                          | disru<br>pted             | 0.7075812<br>27436823<br>1      | True                        | Immun<br>oglobu<br>lin do<br>main; I<br>mmun<br>oglobu<br>lin<br>t-set d<br>omain                                                                         |                      | Protein<br>tyrosin<br>e and s<br>erine/th<br>reonine<br>kinase | FGFR2 (NM_000141) -<br>ADAM12 (NM_003474)<br>rearrangement: c.2026:FGF<br>R2_c.1618+5260:ADAM12d<br>up   Protein Fusion: mid-exon<br>{FGFR2:ADAM12}                                                                                    | NA                                   |
| EVT-P-002<br>0164-T01-I<br>M6-28 | P-0020<br>164-T0<br>1-IM6 |                            | FGFR2<br>-ADAM<br>9  | ADA<br>M9                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>738              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>t-set d<br>omain |                      |                                                                | FGFR2 (NM_000141) -<br>ADAM9 (NM_003816)<br>fusion: t(8;10(p11.22;q26.13)<br>(chr8:g.38901161::chr10:g.1<br>23242738)   Protein Fusion:<br>in frame {FGFR2:ADAM9}                                                                      | NA                                   |
| EVT-P-001<br>8920-T01-I<br>M6-29 | P-0018<br>920-T0<br>1-IM6 |                            | FGFR2<br>-AHCY<br>L1 | AH<br>CYL<br>1           | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>706              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>t-set d<br>omain |                      |                                                                | FGFR2 (NM_000141) -<br>AHCYL1 (NM_006621)<br>fusion (FGFR2 exons 1-17<br>fused in-frame with AHCYL1<br>exons 2-17): t(1;10)(p13.3;q2<br>6.13)(chr1:g.110531185::chr<br>10:g.123242706)   Protein<br>Fusion: in frame<br>{FGFR2:AHCYL1} | NA                                   |

| event_id                         | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name      | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains                                                                                                                                                            | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                          | source_site<br>2_effect_on<br>_frame |
|----------------------------------|---------------------------|----------------------------|----------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-000<br>0644-T01-I<br>M3-30 | P-0000<br>644-T0<br>1-IM3 |                            | FGFR2<br>-AHCY<br>L1 | AH<br>CYL<br>1           | in-fr<br>ame                 | thre<br>_pr<br>ime      | 123239<br>745              | 18                          | 768                                 | True                           | lost                      | 0.0                              | False                       |                                                                                                                                                           | Prote<br>in tyr<br>osine<br>and<br>serin<br>e/thr<br>eoni<br>ne ki<br>nase<br>; Im<br>mun<br>oglo<br>bulin<br>dom<br>ain; I<br>mmu<br>nogl<br>obuli<br>n<br>I-set<br>dom<br>ain |                           | NA   Protein fusion: in frame<br>(AHCYL1-FGFR2)                                                                                                                                 | NA                                   |
| EVT-P-001<br>2666-T01-I<br>M5-31 | P-0012<br>666-T0<br>1-IM5 |                            | FGFR2<br>-AMOT       | AM<br>OT                 | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>762              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                                 |                           | FGFR2 (NM_000141) -<br>AMOT (NM_001113490)<br>Rearrangement : t{(10;X)(q26<br>.13.1;q23)(chr10:g.12324176<br>2::chrX:g.112058354)  <br>Protein fusion: in frame<br>(FGFR2-AMOT) | NA                                   |
| EVT-P-004<br>7504-T01-I<br>M6-32 | P-0047<br>504-T0<br>1-IM6 |                            | FGFR2<br>-ATAD<br>1  | ATA<br>D1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>680              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                                 |                           | FGFR2 (NM_000141) -<br>ATAD1 (NM_032810) fusion:<br>c.2301+1532:FGFR2_c.831+<br>3993:ATAD1del   Protein<br>Fusion: in frame<br>{FGFR2:ATAD1}                                    | NA                                   |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                 | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0017424-T01-IM6-33 | P-0017424-T01-IM6 |            | FGFR2-ATE1  | ATE1         | out-of-frame | five_prime  | 123240443          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |                                              |                   | FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.2302-908:FGFR2_c.1379-520:ATE1dup   Protein Fusion: out of frame {FGFR2:ATE1} | NA                           |
| EVT-P-0063512-T01-IM7-35 | P-0063512-T01-IM7 |            | FGFR2-ATE1  | ATE1         | unknown      | five_prime  | 123263336          | 10              | 469                         | False                  | lost          | 0.0                      | False               | Immunoglobulin domain; Immunoglobulin I-set domain                                                | Protein tyrosine and serine/threonine kinase |                   | FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.1407:FGFR2_c.943-1893:ATE1dup   Protein Fusion: mid-exon {FGFR2:ATE1}         | NA                           |
| EVT-P-0063625-T01-IM7-36 | P-0063625-T01-IM7 |            | FGFR2-ATE1  | ATE1         | in-frame     | five_prime  | 123242847          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |                                              |                   | FGFR2 (NM_000141) - ATE1 (NM_001001976) fusion: c.2301+365:FGFR2_c.975+6106:ATE1dup   Protein Fusion: in frame {FGFR2:ATE1}            | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                     | source_site_2_effect_on_frame |
|--------------------------|-------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0052794-T01-IM6-38 | P-0052794-T01-IM6 |            | FGFR2-BAIAP2L1 | BAIAP2L1     | in-frame     | five_prime  | 123240813          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::chr10:g.123240813)   Protein Fusion: in frame {FGFR2:BAIAP2L1}                     | NA                            |
| EVT-P-0059296-T01-IM7-39 | P-0059296-T01-IM7 |            | FGFR2-BAIAP2L1 | BAIAP2L1     | in-frame     | five_prime  | 123242027          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98029133::chr10:g.123242027)   Protein Fusion: in frame {FGFR2:BAIAP2L1}                     | NA                            |
| EVT-P-0004520-T01-IM5-41 | P-0004520-T01-IM5 |            | FGFR2-BICC1    | BICC1        | in-frame     | five_prime  | 123240269          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2(NM_000141) - BICC1( NM_001080512) Fusion (FGFR2 exon 17 fused with exon 10 of BICC1) : c.2302-734:FGFR2_c.1180-924:BICC1inv   Protein fusion: in frame (FGFR2-BICC1) | NA                            |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status | tar_get_role           | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_truncated | retain_ed_domains                                                                                                                                         | lost_domains | disrupt_ed_domains | source_annotation_text                                                                                                                                                                       | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|--------------|-----------------|---------------|---------------|------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>3174-T01-I<br>M5-42 | P-0013<br>174-T0<br>1-IM5 |              | FGFR2<br>-BICC1 | BIC<br>C1     | in-fr<br>ame  | fiv<br>e_p<br>ri<br>me | 123240<br>123         | 17               | 767                          | True                     | retai<br>ned   | 1.0                        | False               | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |              |                    | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>fusion (FGFR2 exons 1-17<br>with BICC1 exons 16-21): c.<br>2302-588:FGFR2_c.2182-28<br>4:BICC1inv   Protein fusion:<br>in frame (FGFR2-BICC1) | NA                            |
| EVT-P-004<br>9135-T07-I<br>M7-43 | P-0049<br>135-T0<br>7-IM7 |              | FGFR2<br>-BICC1 | BIC<br>C1     | in-fr<br>ame  | fiv<br>e_p<br>ri<br>me | 123239<br>791         | 17               | 767                          | True                     | retai<br>ned   | 1.0                        | False               | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |              |                    | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>fusion: c.2302-256:FGFR2_c<br>.238-36290:BICC1inv  <br>Protein Fusion: in frame<br>{FGFR2:BICC1}                                              | NA                            |
| EVT-P-006<br>8640-T01-I<br>M7-44 | P-0068<br>640-T0<br>1-IM7 |              | FGFR2<br>-BICC1 | BIC<br>C1     | in-fr<br>ame  | fiv<br>e_p<br>ri<br>me | 123240<br>444         | 17               | 767                          | True                     | retai<br>ned   | 1.0                        | False               | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |              |                    | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>fusion: c.2302-909:FGFR2_c<br>.238-18504:BICC1inv  <br>Protein Fusion: in frame<br>{FGFR2:BICC1}                                              | NA                            |

| event_id                         | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                  | source_site<br>2_effect_on<br>_frame |
|----------------------------------|---------------------------|----------------------------|-----------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-004<br>9135-T03-I<br>M6-45 | P-0049<br>135-T0<br>3-IM6 |                            | FGFR2<br>-BICC1 | BIC<br>C1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123239<br>791              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>fusion: c.2302-256:FGFR2_c<br>.238-36290:BICC1inv  <br>Protein Fusion: in frame<br>{FGFR2:BICC1}         | NA                                   |
| EVT-P-004<br>9133-T04-I<br>M6-46 | P-0049<br>133-T0<br>4-IM6 |                            | FGFR2<br>-BICC1 | BIC<br>C1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>098              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>fusion: c.2301+1114:FGFR2<br>_c.237+29879:BICC1inv  <br>Protein Fusion: in frame<br>{FGFR2:BICC1}        | NA                                   |
| EVT-P-000<br>6885-T01-I<br>M5-48 | P-0006<br>885-T0<br>1-IM5 |                            | FGFR2<br>-BICC1 | BIC<br>C1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>587              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>rearrangement: c.2302-1052:<br>FGFR2_c.238-5239:BICC1in<br>v   Protein fusion: in frame<br>(FGFR2-BICC1) | NA                                   |

| event_id                         | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                           | source_site<br>2_effect_on<br>_frame |
|----------------------------------|---------------------------|----------------------------|-----------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-005<br>1773-T01-I<br>M6-49 | P-0051<br>773-T0<br>1-IM6 |                            | FGFR2<br>-BICC1 | BIC<br>C1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>828              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>fusion: c.2301+384:FGFR2_<br>c.237+13961:BICC1inv  <br>Protein Fusion: in frame<br>{FGFR2:BICC1}  | NA                                   |
| EVT-P-004<br>9133-T02-I<br>M6-50 | P-0049<br>133-T0<br>2-IM6 |                            | FGFR2<br>-BICC1 | BIC<br>C1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>098              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>fusion: c.2301+1114:FGFR2_<br>c.237+29879:BICC1inv  <br>Protein Fusion: in frame<br>{FGFR2:BICC1} | NA                                   |
| EVT-P-005<br>3822-T01-I<br>M6-51 | P-0053<br>822-T0<br>1-IM6 |                            | FGFR2<br>-BICC1 | BIC<br>C1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123243<br>033              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>BICC1 (NM_001080512)<br>fusion: c.2301+179:FGFR2_<br>c.238-1932:BICC1inv  <br>Protein Fusion: in frame<br>{FGFR2:BICC1}   | NA                                   |

| event_id                     | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                           | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>1327-T01-IM6-52 | P-0021<br>327-T01-IM6 |            | FGFR2-BICC1 | BICC1        | in-frame     | five_prime  | 123240344          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21): c.2302-1613:FGFR2_c.237+32254:BICC1inv   Protein Fusion: in frame {FGFR2:BICC1} | NA                           |
| EVT-P-006<br>0245-T01-IM7-53 | P-0060<br>245-T01-IM7 |            | FGFR2-BICC1 | BICC1        | unknown      | five_prime  | 123239377          | 18              | 820                         | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2460:FGFR2_c.1179+1103:BICC1inv   Protein Fusion: mid-exon {FGFR2:BICC1}                                                      | NA                           |
| EVT-P-006<br>8033-T01-IM7-54 | P-0068<br>033-T01-IM7 |            | FGFR2-BICC1 | BICC1        | in-frame     | five_prime  | 123239673          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-138:FGFR2_c.238-40422:BICC1inv   Protein Fusion: in frame {FGFR2:BICC1}                                                  | NA                           |

| event_id                     | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                           | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>9087-T01-IM6-55 | P-0039<br>087-T01-IM6 |            | FGFR2-BICC1 | BICC1        | in-frame     | five_prime  | 123242059          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2301+1153:FGFR2_c.2181+133:BICC1inv   Protein Fusion: in frame {FGFR2:BICC1} | NA                           |
| EVT-P-005<br>4191-T01-IM6-56 | P-0054<br>191-T01-IM6 |            | FGFR2-BICD1 | BICD1        | in-frame     | five_prime  | 123242849          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BICD1 (NM_001714) fusion: t(10;12)(q26.13;p11.21)(chr10:g.123242849::chr12:g.32465127)   Protein Fusion: in frame {FGFR2:BICD1}                              | NA                           |
| EVT-P-006<br>1764-T01-IM7-57 | P-0061<br>764-T01-IM7 |            | FGFR2-BNIP2 | BNIP2        | in-frame     | five_prime  | 123243170          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BNIP2 (NM_004330) fusion: t(10;15)(q26.13;q22.2)(chr10:g.123243170::chr15:g.59975666)   Protein Fusion: in frame {FGFR2:BNIP2}                               | NA                           |

| event_id                     | sample_id             | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>0135-T01-IM6-58 | P-0020<br>135-T01-IM6 |            | FGFR2-BTBD16  | BTBD16       | out-of-frame | five_prime  | 123239702          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - BTBD16 (NM_144587) rearrangement: c.2302-167: FGFR2_c.386-107: BTBD16inv   Protein Fusion: out of frame {FGFR2:BTBD16}            | NA                           |
| EVT-P-003<br>3652-T01-IM6-59 | P-0033<br>652-T01-IM6 |            | FGFR2-CCDC186 | CCDC186      | in-frame     | five_prime  | 123240830          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - CCDC186(NM_018017) Rearrangement : c.2302-1295:FGFR2_c.1222-3356:CCDC186del   Protein Fusion: in frame {FGFR2:CCDC186}            | NA                           |
| EVT-P-006<br>0582-T01-IM7-60 | P-0060<br>582-T01-IM7 |            | FGFR2-CCDC73  | CCDC73       | in-frame     | five_prime  | 123241248          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - CCDC73 (NM_001008391) fusion: t(10;11)(q26.13;p13)(chr10:g.123241248::chr11:g.32747346)   Protein Fusion: in frame {FGFR2:CCDC73} | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                 | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>7574-T01-IM6-61  | P-0037<br>574-T01-IM6 |            | FGFR2-CTNNA3 | CTNNA3       | in-frame     | five_prime  | 123240911          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - CTNNA3 (NM_013266) fusion: c.2302-1376:FGFR2_c.1884+30653:CTNNA3del   Protein Fusion: in frame {FGFR2:CTNNA3}                      | NA                           |
| EVT-P-004<br>4768-T01-IM6-63  | P-0044<br>768-T01-IM6 |            | FGFR2-DIABLO | DIABLO       | in-frame     | five_prime  | 123243152          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - DIABLO (NM_019887) fusion: t(10;12)(q26.13;q24.31)(chr10:g.123243152::chr12:g.122708063)   Protein Fusion: in frame {FGFR2:DIABLO} | NA                           |
| EVT-P-005<br>8539-T01-IM6-101 | P-0058<br>539-T01-IM6 |            | FGFR2-FOXP1  | FOXP1        | unknown      | five_prime  | 123239434          | 18              | 801                         | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - FOXP1 (NM_001244814) fusion: t(3;10)(p13;q26.13)(chr3:g.71113500::chr10:g.123239434)   Protein Fusion: mid-exon {FGFR2:FOXP1}      | NA                           |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name       | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains                                                                                                              | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                             | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|-----------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-006<br>1783-T01-I<br>M7-102 | P-0061<br>783-T0<br>1-IM7 |                            | FGFR2<br>-FOXP<br>1   | FOX<br>P1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>483              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                   |                           | FGFR2 (NM_000141) -<br>FOXP1 (NM_001244814)<br>fusion: t(3;10)(p13;q26.13)(c<br>hr3:g.71192573::chr10:g.123<br>241483)   Protein Fusion: in<br>frame {FGFR2:FOXP1} | NA                                   |
| EVT-P-005<br>5215-T01-I<br>M6-103 | P-0055<br>215-T0<br>1-IM6 |                            | FGFR2<br>-GRID<br>1   | GRI<br>D1                | out-of-fr<br>ame             | fiv<br>e_<br>pri<br>me  | 123323<br>934              | 4                           | 152                                 | True                           | lost                      | 0.0                              | False                       | Immun<br>oglobu<br>lin do<br>main                                                                                                                         | Prote<br>in tyr<br>osine<br>and<br>serin<br>e/thr<br>eoni<br>ne ki<br>nase<br>; Im<br>mun<br>oglo<br>bulin<br>I-set<br>dom<br>ain |                           | FGFR2 (NM_000141) -<br>GRID1 (NM_017551)<br>rearrangement: c.454+82:FG<br>FR2_c.726+103661:GRID1d<br>el   Protein Fusion: out of<br>frame {FGFR2:GRID1}            | NA                                   |
| EVT-P-001<br>5437-T01-I<br>M6-104 | P-0015<br>437-T0<br>1-IM6 |                            | FGFR2<br>-HSPA<br>12A | HSP<br>A12<br>A          | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>591              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                   |                           | FGFR2 (NM_000141) -<br>HSPA12A (NM_025015)<br>rearrangement: c.2301+621:<br>FGFR2_c.663+3346:HSPA1<br>2Adel   Protein fusion: in<br>frame (FGFR2-HSPA12A)          | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name        | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                          | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|------------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-000<br>9701-T01-I<br>M5-105 | P-0009<br>701-T0<br>1-IM5 |                            | FGFR2<br>-KCNH<br>7    | KC<br>NH7                | unk<br>now<br>n              | fiv<br>e_<br>pri<br>me  | 123239<br>444              | 18                          | 798                                 | False                          | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>KCNH7 (NM_033272)<br>Rearrangement :<br>t(2;10)(q24.3; q26.13) (chr2:<br>g.163263969::chr10:g.12323<br>9444)   Protein fusion:<br>mid-exon (FGFR2-KCNH7) | NA                                   |
| EVT-P-000<br>8658-T01-I<br>M5-107 | P-0008<br>658-T0<br>1-IM5 |                            | FGFR2<br>-KIAA1<br>217 | KIA<br>A12<br>17         | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>193              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>KIAA1217 (NM_019590)<br>Rearrangement: c.2302-165<br>8:FGFR2_c.355-22274:KIAA<br>1217inv   Protein fusion: in<br>frame (FGFR2-KIAA1217)                  | NA                                   |
| EVT-P-003<br>4784-T01-I<br>M6-108 | P-0034<br>784-T0<br>1-IM6 |                            | FGFR2<br>-KIAA1<br>217 | KIA<br>A12<br>17         | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>254              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>KIAA1217 (NM_019590)<br>fusion: c.2301+958:FGFR2_<br>c.355-49659:KIAA1217inv  <br>Protein Fusion: in frame<br>{FGFR2:KIAA1217}                           | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name        | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains                                                                                                                                                            | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                  | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|------------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-001<br>3405-T01-I<br>M5-109 | P-0013<br>405-T0<br>1-IM5 |                            | FGFR2<br>-KIAA1<br>217 | KIA<br>A12<br>17         | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>766              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                                 |                           | FGFR2 (NM_000141) -<br>KIAA1217 (NM_019590)<br>Rearrangement: c.2301+144<br>6:FGFR2_c.354+73179:KIA<br>A1217inv   Protein fusion: in<br>frame (FGFR2-KIAA1217)          | NA                                   |
| EVT-P-000<br>3660-T01-I<br>M5-110 | P-0003<br>660-T0<br>1-IM5 |                            | FGFR2<br>-KIF14        | KIF<br>14                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>038              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                                 |                           | t(1;10)(q32.1;q26.13)(chr1:g.<br>200555931::chr10:g.123240<br>038)   Protein fusion: in<br>frame (FGFR2-KIF14)                                                          | NA                                   |
| EVT-P-001<br>3241-T01-I<br>M5-111 | P-0013<br>241-T0<br>1-IM5 |                            | FGFR2<br>-KLHL<br>29   | KLH<br>L29               | unk<br>now<br>n              | thre<br>e_<br>pri<br>me | 123241<br>923              | 18                          | 768                                 | True                           | lost                      | 0.0                             | False                       |                                                                                                                                                           | Prote<br>in tyr<br>osine<br>and<br>serin<br>e/thr<br>eoni<br>ne ki<br>nase<br>; Im<br>mun<br>oglo<br>bulin<br>dom<br>ain; I<br>mmu<br>nogl<br>obuli<br>n<br>I-set<br>dom<br>ain |                           | KLHL29 (NM_052920) -<br>FGFR2 (NM_000141)<br>rearrangement: t(2;10)(p24.1<br>;q26.13)(chr2:g.:23647979::c<br>hr10:g.123241923)  <br>Transcript fusion<br>(KLHL29-FGFR2) | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name       | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                          | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|-----------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-001<br>9439-T02-I<br>M6-112 | P-0019<br>439-T0<br>2-IM6 |                            | FGFR2<br>-L3MB<br>TL3 | L3M<br>BTL<br>3          | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123239<br>680              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>L3MBTL3 (NM_032438)<br>rearrangement: t(6;10)(q23.1<br>;q26.1)(chr6:g.130408449::c<br>hr10:g.123239680)   Protein<br>Fusion: in frame<br>{FGFR2:L3MBTL3} | NA                                   |
| EVT-P-001<br>1394-T01-I<br>M5-113 | P-0011<br>394-T0<br>1-IM5 |                            | FGFR2<br>-LAMC<br>1   | LAM<br>C1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>042              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>LAMC1 (NM_002293)<br>rearrangement: t(1;10)(q25.3;<br>q26.13)(chr1:g.183109471::c<br>hr10:g.123240042 )   Protein<br>fusion: in frame<br>(FGFR2-LAMC1)   | NA                                   |
| EVT-P-006<br>4855-T01-I<br>M7-115 | P-0064<br>855-T0<br>1-IM7 |                            | FGFR2<br>-MBIP        | MBI<br>P                 | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>070              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) - MBIP<br>(NM_016586) fusion: t(10;14<br>) (q26.13;q13.3)(chr10:g.123<br>240070::chr14:g.36787130)  <br>Protein Fusion: in frame<br>{FGFR2:MBIP}              | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name     | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                        | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|---------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-003<br>0844-T01-I<br>M6-116 | P-0030<br>844-T0<br>1-IM6 |                            | FGFR2<br>-MYH1<br>5 | MY<br>H15                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>665              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>MYH15 (NM_014981)<br>rearrangement: t(3;10)(q13.1<br>3;q26.13)(chr3:g.108134575:<br>:chr10:g.123240665)  <br>Protein Fusion: in frame<br>{FGFR2:MYH15} | NA                                   |
| EVT-P-002<br>7623-T01-I<br>M6-117 | P-0027<br>623-T0<br>1-IM6 |                            | FGFR2<br>-MYLK      | MYL<br>K                 | unk<br>now<br>n              | fiv<br>e_<br>pri<br>me  | 123240<br>959              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141)<br>rearrangement: t(3;10)(q21.1<br>;q26.13)(chr3:g.123525612::<br>chr10:g.123240959)  <br>Transcript Fusion<br>{FGFR2:MYLK}                                 | NA                                   |
| EVT-P-003<br>8235-T01-I<br>M6-118 | P-0038<br>235-T0<br>1-IM6 |                            | FGFR2<br>-NOL4      | NOL<br>4                 | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>418              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>NOL4 (NM_003787) fusion: t<br>(10;18)(q26.13;q12.1)(chr10:<br>g.123240418::chr18:g.31580<br>817)   Protein Fusion: in<br>frame {FGFR2:NOL4}            | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name     | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                                                                         | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|---------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-001<br>2423-T02-I<br>M6-119 | P-0012<br>423-T0<br>2-IM6 |                            | FGFR2<br>-NOL4      | NOL<br>4                 | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>055              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>NOL4 (NM_003787) fusion<br>(FGFR2 exons 1-17 fused<br>in-frame with NOL4 exons<br>7-11): t(10;18)(q26.13;q12.1)<br>(chr10:g.123242055::chr18:g<br>.31563418)   Protein Fusion:<br>in frame {FGFR2:NOL4} | NA                                   |
| EVT-P-000<br>6328-T02-I<br>M6-120 | P-0006<br>328-T0<br>2-IM6 |                            | FGFR2<br>-NRBF<br>2 | NR<br>BF2                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>246              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>NRBF2 (NM_030759) fusion:<br>c.2302-1711:FGFR2_c.157-<br>363NRBF2inv   Protein<br>Fusion: in frame<br>{FGFR2:NRBF2}                                                                                     | NA                                   |
| EVT-P-005<br>8339-T01-I<br>M6-121 | P-0058<br>339-T0<br>1-IM6 |                            | FGFR2<br>-PAH       | PAH                      | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>026              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) - PAH<br>(NM_000277) fusion: t(10;12<br>) (q26.13;q23.2)(chr10:g.123<br>240026::chr12:g.103294473)<br>  Protein Fusion: in frame<br>{FGFR2:PAH}                                                              | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name     | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                         | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|---------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-001<br>5500-T01-I<br>M6-122 | P-0015<br>500-T0<br>1-IM6 |                            | FGFR2<br>-PAL<br>LD | PAL<br>LD                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123239<br>869              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>PALLD (NM_001166108)<br>Rearrangement : t(4;10)(q33,<br>q26.13)(chr4:g.169605121::c<br>hr10:g.123239869)   Protein<br>fusion: in frame<br>(FGFR2-PALLD) | NA                                   |
| EVT-P-002<br>1448-T03-I<br>M6-123 | P-0021<br>448-T0<br>3-IM6 |                            | FGFR2<br>-PKP4      | PKP<br>4                 | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>661              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>PKP4 (NM_003628)<br>rearrangement: t(2;10)(q24.1<br>;q26.13)(chr2:g.159395413::<br>chr10:g.123242661)   Protein<br>Fusion: in frame<br>{FGFR2:PKP4}     | NA                                   |
| EVT-P-005<br>9669-T01-I<br>M7-125 | P-0059<br>669-T0<br>1-IM7 |                            | FGFR2<br>-POC1<br>B | PO<br>C1B                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>798              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>POC1B (NM_001199777.2)<br>fusion: t(10;12)(q26.13;q21.3<br>3)(chr10:g.123242798::chr12<br>:g.89857934)   Protein<br>Fusion: in frame<br>{FGFR2:POC1B}   | NA                                   |

| event_id                  | sample_id         | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                   | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                         | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------------------------------------------------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0008657-T01-IM5-126 | P-0008657-T01-IM5 |            | FGFR2-PPP1R21  | PPP1R21      | in-frame     | five_prime  | 123240684          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain |              |                   | FGFR1 (NM_000141) - PPP1R21 (NM_152994) Rearrangement : t(2;10)(2p16.3;10q26.13)(chr2:g.48721864::chr10:g.123240684)   Protein fusion: in frame (FGFR2-KLRAQ1) | NA                           |
| EVT-P-0057247-T02-IM7-127 | P-0057247-T02-IM7 |            | FGFR2-PRDX3    | PRDX3        | in-frame     | five_prime  | 123242894          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - PRDX3 (NM_006793) fusion: c.2301+318:FGFR2_c.36+287:PRDX3del   Protein Fusion: in frame {FGFR2:PRDX3}                                      | NA                           |
| EVT-P-0011394-T01-IM5-129 | P-0011394-T01-IM5 |            | FGFR2-RABGAP1L | RABGAP1L     | in-frame     | five_prime  | 123240065          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - RABGAP1L (NM_014857) rearrangement: t(1;10)(q25.1;q26.13)(chr1:g.174863117::chr10:g.123240065)   Protein fusion: in frame (FGFR2-RABGAP1L) | NA                           |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name        | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                             | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|------------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-001<br>2639-T01-I<br>M5-130 | P-0012<br>639-T0<br>1-IM5 |                            | FGFR2<br>-SEPTI<br>N10 | SEPTI<br>N10             | unk<br>now<br>n              | fiv<br>e_<br>pri<br>me  | 123240<br>798              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>SEPT10 (NM_144710)<br>rearrangement: t(2;10)(q13;q<br>26.13)(chr2:g.110325395::chr<br>10:g.123240798)   Protein<br>fusion: mid-exon<br>(FGFR2-SEPT10)       | NA                                   |
| EVT-P-005<br>5202-T01-I<br>M6-131 | P-0055<br>202-T0<br>1-IM6 |                            | FGFR2<br>-SH3G<br>LB1  | SH3<br>GLB<br>1          | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>019              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>SH3GLB1 (NM_001206651)<br>rearrangement: t(1;10)(p22.3<br>;q26.13)(chr1:g.87191292::c<br>hr10:g.123242019)   Protein<br>Fusion: in frame<br>{FGFR2:SH3GLB1} | NA                                   |
| EVT-P-001<br>1516-T01-I<br>M5-132 | P-0011<br>516-T0<br>1-IM5 |                            | FGFR2<br>-SHC2         | SH<br>C2                 | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123243<br>182              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>SH2C (NM_012435)<br>Rearrangement : t(10;19)(q2<br>6.13;p13.3)(chr10:g.1232431<br>82::chr19:g.447779)   Protein<br>fusion: in frame<br>(FGFR2-SHC2)         | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name     | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                        | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|---------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-003<br>2152-T01-I<br>M6-133 | P-0032<br>152-T0<br>1-IM6 |                            | FGFR2<br>-SHTN<br>1 | SHT<br>N1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>513              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>KIAA1598 (NM_001127211)<br>fusion: c.2301+1699:FGFR2<br>_c.534+751:KIAA1598  <br>Protein Fusion: in frame<br>{FGFR2:KIAA1598}          | NA                                   |
| EVT-P-005<br>7005-T01-I<br>M6-134 | P-0057<br>005-T0<br>1-IM6 |                            | FGFR2<br>-SHTN<br>1 | SHT<br>N1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123239<br>901              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>KIAA1598 (NM_001127211)<br>fusion: c.2302-366:FGFR2_c<br>.534+660:KIAA1598del  <br>Protein Fusion: in frame<br>{FGFR2:KIAA1598}        | NA                                   |
| EVT-P-006<br>4573-T01-I<br>M7-135 | P-0064<br>573-T0<br>1-IM7 |                            | FGFR2<br>-SKI       | SKI                      | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123242<br>232              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                             | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) - SKI<br>(NM_003036) fusion: t(1;10)(<br>p36.33;q26.13)(chr1:g.22078<br>11::chr10:g.123242232)  <br>Protein Fusion: in frame<br>{FGFR2:SKI} | NA                                   |

| event_id                  | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intro_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                      | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|---------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0054244-T01-IM6-136 | P-0054244-T01-IM6 |            | FGFR2-SORBS1 | SORBS1       | in-frame     | five_prime  | 123243201          | 17              | 767                         | True                | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - SORBS1 (NM_001034954) fusion: c.2301+11:FGFR2_c.1738-382:SORBS1del   Protein Fusion: in frame {FGFR2:SORBS1}                            | NA                           |
| EVT-P-0057277-T01-IM6-137 | P-0057277-T01-IM6 |            | FGFR2-SRRT   | SRRT         | out-of-frame | five_prime  | 123242651          | 17              | 767                         | True                | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - SRRT (NM_015908) fusion: t(7;10)(q22.1;q26.13)(chr7:g.100475766::chr10:g.123242651)   Protein Fusion: out of frame {FGFR2:SRRT}         | NA                           |
| EVT-P-0055975-T01-IM6-139 | P-0055975-T01-IM6 |            | FGFR2-STAT4  | STAT4        | in-frame     | five_prime  | 123240312          | 17              | 767                         | True                | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - STAT4 (NM_001243835) rearrangement: t(2;10)(q32.3;q26.13)(chr2:g.191936121::chr10:g.123240312)   Protein Fusion: in frame {FGFR2:STAT4} | NA                           |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name     | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                       | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|---------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-004<br>5185-T01-I<br>M6-141 | P-0045<br>185-T0<br>1-IM6 |                            | FGFR2<br>-TACC<br>2 | TAC<br>C2                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>014              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>TACC2 (NM_206862) fusion:<br>c.2302-479:FGFR2_c.8128-<br>2249:TACC2inv   Protein<br>Fusion: in frame<br>{FGFR2:TACC2} | NA                                   |
| EVT-P-003<br>1242-T01-I<br>M6-142 | P-0031<br>242-T0<br>1-IM6 |                            | FGFR2<br>-TACC<br>2 | TAC<br>C2                | unk<br>now<br>n              | fiv<br>e_<br>pri<br>me  | 123239<br>432              | 18                          | 802                                 | False                          | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>TACC2 (NM_206862) fusion:<br>c.2405:FGFR2_c.7896+237:<br>TACC2inv   Protein Fusion:<br>mid-exon {FGFR2:TACC2}         | NA                                   |
| EVT-P-002<br>3364-T03-I<br>M6-143 | P-0023<br>364-T0<br>3-IM6 |                            | FGFR2<br>-TACC<br>2 | TAC<br>C2                | unk<br>now<br>n              | fiv<br>e_<br>pri<br>me  | 123241<br>587              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>TACC2 (NM_206862)<br>rearrangement: c.2301+1625<br>:FGFR2_c.34-5469:TACC2d<br>up   Antisense Fusion                   | NA                                   |

| event_id                          | sample_id                 | patient_id | fusion_name         | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                      | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|---------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>4187-T01-I<br>M6-144 | P-0024<br>187-T0<br>1-IM6 |            | FGFR2<br>-TACC<br>2 | TAC<br>C2    | unknown      | five_prime  | 123240<br>573      | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - TACC2 (NM_206862) Fusion (FGFR2 exon 17 with TACC2 exon12) : c.2302-1038:FGFR2_c.8009:TACC2inv   Protein Fusion: mid-exon {FGFR2:TACC2} | NA                           |
| EVT-P-005<br>0962-T01-I<br>M6-145 | P-0050<br>962-T0<br>1-IM6 |            | FGFR2<br>-TACC<br>2 | TAC<br>C2    | in-frame     | five_prime  | 123241<br>844      | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1368:FGFR2_c.7608+332:TACC2inv   Protein Fusion: in frame {FGFR2:TACC2}                                | NA                           |
| EVT-P-003<br>6358-T01-I<br>M6-146 | P-0036<br>358-T0<br>1-IM6 |            | FGFR2<br>-TACC<br>2 | TAC<br>C2    | in-frame     | five_prime  | 123239<br>743      | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2302-208:FGFR2_c.7344+468:TACC2inv   Protein Fusion: in frame {FGFR2:TACC2}                                 | NA                           |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name      | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                    | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|----------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-005<br>7746-T01-I<br>M6-149 | P-0057<br>746-T0<br>1-IM6 |                            | FGFR2<br>-TACC<br>2  | TAC<br>C2                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>998              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>TACC2 (NM_206862) fusion:<br>c.2301+1214:FGFR2_c.8128<br>-402:TACC2inv   Protein<br>Fusion: in frame<br>{FGFR2:TACC2}                              | NA                                   |
| EVT-P-005<br>7746-T02-I<br>M6-150 | P-0057<br>746-T0<br>2-IM6 |                            | FGFR2<br>-TACC<br>2  | TAC<br>C2                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>998              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>TACC2 (NM_206862) fusion:<br>c.2301+1214:FGFR2_c.8128<br>-402:TACC2inv   Protein<br>Fusion: in frame<br>{FGFR2:TACC2}                              | NA                                   |
| EVT-P-002<br>2337-T06-I<br>M6-151 | P-0022<br>337-T0<br>6-IM6 |                            | FGFR2<br>-TBC1<br>D4 | TBC<br>1D4               | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>723              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                      |                           | FGFR2 (NM_000141) -<br>TBC1D4 (NM_014832)<br>fusion: t(10;13)(q26.13;q22.2<br>) (chr10:g.123241723::chr13:<br>g.75946350)   Protein<br>Fusion: in frame<br>{FGFR2:TBC1D4} | NA                                   |

| event_id                      | sample_id             | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intro_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                        | source_site_2_effect_on_frame |
|-------------------------------|-----------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|---------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>2337-T04-IM6-152 | P-0022<br>337-T04-IM6 |            | FGFR2-TBC1D4 | TBC1D4       | in-frame     | five_prime  | 123241723          | 17              | 767                         | True                | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946350)   Protein Fusion: in frame {FGFR2:TBC1D4}          | NA                            |
| EVT-P-000<br>5291-T01-IM5-153 | P-0005<br>291-T01-IM5 |            | FGFR2-TFEC   | TFEC         | unknown      | five_prime  | 123241729          | 17              | 767                         | True                | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_022970) - TFEC (NM_012252) Rearrangement: t(10;7)(q26.13;q31.2)(chr10:g.123241729;chr7:g.115644112)   Transcript fusion (FGFR2-TFEC)                | NA                            |
| EVT-P-001<br>6640-T01-IM6-154 | P-0016<br>640-T01-IM6 |            | FGFR2-TTC28  | TTC28        | in-frame     | five_prime  | 123241944          | 17              | 767                         | True                | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000142) - TTC28 (NM_001145418) Rearrangement : t(10;22)(q26.11;q12.1)(chr10:g.123241944::chr22:g.28666001)   Protein Fusion: in frame {FGFR2:TTC28} | NA                            |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0051558-T01-IM6-155 | P-0051558-T01-IM6 |            | FGFR2-TTC28 | TTC28        | in-frame     | five_prime  | 123243158          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - TTC28 (NM_001145418) fusion: t(10;22)(q26.13;q12.1)(chr10:g.123243158::chr22:g.28605071)   Protein Fusion: in frame {FGFR2:TTC28} | NA                           |
| EVT-P-0029726-T01-IM6-156 | P-0029726-T01-IM6 |            | FGFR2-UBP1  | UBP1         | in-frame     | five_prime  | 123240062          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - UBP1 (NM_001128161) fusion: t(3;10)(p22.3;q26.13)(chr3:g.33449450::chr10:g.123240062)   Protein Fusion: in frame {FGFR2:UBP1}     | NA                           |
| EVT-P-0004089-T01-IM5-158 | P-0004089-T01-IM5 |            | FGFR2-VCL   | VCL          | unknown      | five_prime  | 123242749          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2(NM_000141) - VCL (NM_014000) Rearrangement : c.2301+463:FGFR2_c.1032:VCLinv   Protein fusion: mid-exon (FGFR2-VCL)                              | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name      | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains                                                                                      | disrupted_domains | source_annotation_text                                                                                                                                    | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-005<br>9238-T02-I<br>M7-159 | P-0059<br>238-T0<br>2-IM7 |            | FGFR2<br>-WAC    | WAC          | in-frame     | five_prime  | 123242066          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |                                                                                                   |                   | FGFR2 (NM_000141) - WAC (NM_016628) fusion: c.2301+1146:FGFR2_c.1437+679:WACinv   Protein Fusion: in frame {FGFR2:WAC}                                    | NA                           |
| EVT-P-005<br>8031-T01-I<br>M6-160 | P-0058<br>031-T0<br>1-IM6 |            | FGFR2<br>-WAC    | WAC          | in-frame     | five_prime  | 123243180          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |                                                                                                   |                   | FGFR2 (NM_000141) - WAC (NM_016628) fusion: c.2301+32:FGFR2_c.382-351:WACinv   Protein Fusion: in frame {FGFR2:WAC}                                       | NA                           |
| EVT-P-002<br>5807-T01-I<br>M6-161 | P-0025<br>807-T0<br>1-IM6 |            | FGFR2<br>-YTHDF3 | YTHDF3       | unknown      | three_prime | 123242317          | 18              | 768                         | True                   | lost          | 0.0                      | False               |                                                                                                   | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |                   | YTHDF3 (NM_152758) - FGFR2 (NM_000141) rearrangement: t(8;10)(q12.3;q26.13)(chr8:g.64081967::chr10:g.123242317)   Protein Fusion: mid-exon {YTHDF3:FGFR2} | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                          | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0056399-T01-IM6-162 | P-0056399-T01-IM6 |            | FGFR2-ZFAT  | ZFAT         | in-frame     | five_prime  | 123240282          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - ZFAT (NM_020863) rearrangement: t(8;10)(q24.2;q26.13)(chr8:g.135564146::chr10:g.123240282)   Protein Fusion: in frame {FGFR2:ZFAT}                                          | NA                           |
| EVT-P-0042241-T01-IM6-163 | P-0042241-T01-IM6 |            | FGFR2-ZMYM4 | ZMYM4        | in-frame     | five_prime  | 123242889          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - ZMYM4 (NM_005095) fusion: t(1;10)(p34.3;q26.13)(chr1:g.35833702::chr10:g.123242889)   Protein Fusion: in frame {FGFR2:ZMYM4}                                                | NA                           |
| EVT-P-0016195-T01-IM6-164 | P-0016195-T01-IM6 |            | GAB2-FGFR2  | GAB2         | in-frame     | five_prime  | 123242812          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - GAB2 (NM_080491) fusion (FGFR2 exons 1-17 fused with GAB2 exons 2-10) : t(10;11)(q26.11;q14.1)(chr10:g.123242812::chr11:g.78030553)   Protein Fusion: in frame {FGFR2:GAB2} | NA                           |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains | lost_<br>dom<br>ains                                                                                                                                                            | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                          | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|-----------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-001<br>3756-T01-I<br>M5-165 | P-0013<br>756-T0<br>1-IM5 |                            | INA-F<br>GFR2   | INA                      | in-fr<br>ame                 | thr<br>ee_<br>pr<br>ime | 123242<br>695              | 18                          | 768                                 | True                           | lost                      | 0.0                              | False                       |                          | Prote<br>in tyr<br>osine<br>and<br>serin<br>e/thr<br>eoni<br>ne ki<br>nase<br>; Im<br>mun<br>oglo<br>bulin<br>dom<br>ain; I<br>mmu<br>nogl<br>obuli<br>n<br>I-set<br>dom<br>ain |                           | INA (NM_032727) - FGFR2<br>(NM_000141)<br>Rearrangement : c.1066-350<br>0:INA_c.2301+517:FGFR2in<br>v   Protein fusion: in frame<br>(INA-FGFR2) | NA                                   |
| EVT-P-001<br>3756-T03-I<br>M6-166 | P-0013<br>756-T0<br>3-IM6 |                            | INA-F<br>GFR2   | INA                      | in-fr<br>ame                 | thr<br>ee_<br>pr<br>ime | 123242<br>695              | 18                          | 768                                 | True                           | lost                      | 0.0                              | False                       |                          | Prote<br>in tyr<br>osine<br>and<br>serin<br>e/thr<br>eoni<br>ne ki<br>nase<br>; Im<br>mun<br>oglo<br>bulin<br>dom<br>ain; I<br>mmu<br>nogl<br>obuli<br>n<br>I-set<br>dom<br>ain |                           | INA (NM_032727) - FGFR2<br>(NM_000141)<br>rearrangement: c.1066-3500:<br>INA_c.2301+517:FGFR2inv  <br>Protein Fusion: in frame<br>{INA:FGFR2}   | NA                                   |

| event_id                  | sample_id         | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                  | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0002810-T01-IM3-169 | P-0002810-T01-IM3 |            | KIAA1217-FGFR2 | KIAA1217     | in-frame     | five_prime  | 123240874          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | NA   Protein fusion: in frame (FGFR2-KIAA1217)                                                                                                          | NA                           |
| EVT-P-000386-T02-IM5-170  | P-0000386-T02-IM5 |            | MARVELD3-FGFR2 | MARVELD3     | unknown      | five_prime  | 123243101          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | MARVELD3 (NM_001017967) - FGFR2 (NM_000141) rearrangement: t(16;10)(16q22.2;10q26.13) (chr16:g.71660650::chr10:123243101)   Antisense fusion            | NA                           |
| EVT-P-0024094-T01-IM6-172 | P-0024094-T01-IM6 |            | MYH9-FGFR2     | MYH9         | in-frame     | five_prime  | 123240418          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - MYH9 (NM_002473) rearrangement: t(10;22)(q26.13;q12.3)(chr10:g.123240418::chr22:g.36682296)   Protein Fusion: in frame {FGFR2:MYH9} | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intro_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                   | lost_domains                                                                                       | disrupted_domains | source_annotation_text                                                                                                                                      | source_site_2_effect_on_frame |
|-------------------------------|-----------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|---------------------|---------------|--------------------------|---------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>2208-T01-IM6-173 | P-0032<br>208-T01-IM6 |            | MYO18B-FGFR2 | MYO18B       | in-frame     | five_prime  | 123241929          | 17              | 767                         | True                | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain |                                                                                                    |                   | FGFR2 (NM_000141) - MYO18B (NM_032608) rearrangement: t(10;22)(q26.13;q12.2)(chr10:g.123241929::chr22:g.26280218)   Protein Fusion: in frame {FGFR2:MYO18B} | NA                            |
| EVT-P-002<br>1975-T01-IM6-175 | P-0021<br>975-T01-IM6 |            | NID2-FGFR2   | NID2         | out-of-frame | three_prime | 123240637          | 18              | 768                         | True                | lost          | 0.0                      | False               |                                                                                                    | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain |                   | NID2 (NM_007361) - FGFR2 (NM_000141) rearrangement: t(10;14)(q26.13;q22.1)(chr10:g.123240637::chr14:g.52482743)   Protein Fusion: out of frame {NID2:FGFR2} | NA                            |
| EVT-P-001<br>2423-T01-IM5-176 | P-0012<br>423-T01-IM5 |            | NOL4-FGFR2   | NOL4         | in-frame     | five_prime  | 123242054          | 17              | 767                         | True                | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain |                                                                                                    |                   | FGFR2 (NM_000141) - NOL4 (NM_003787) Rearrangement :t(10;18)(q26.13;q12.1)(chr10:g.123242054::chr18:g.31563418)   Protein fusion: in frame (FGFR2-NOL4)     | NA                            |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                        | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0023103-T01-IM6-177 | P-0023103-T01-IM6 |            | NOL4-FGFR2  | NOL4         | in-frame     | five_prime  | 123241921          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused to NOL4 exons 10-11): t(10;18)(q26.13;q12.1)(chr10:g.123241921::chr18:g.31518701)   Protein Fusion: in frame {FGFR2:NOL4} | NA                           |
| EVT-P-0008693-T01-IM5-178 | P-0008693-T01-IM5 |            | NRAP-FGFR2  | NRAP         | in-frame     | five_prime  | 123242277          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - NRAP (NM_198060) rearrangement: c.2301+935:FGFR2_c.2458-427:NRAPdel   Protein fusion: in frame (FGFR2-NRAP)                                                               | NA                           |
| EVT-P-0006328-T01-IM5-179 | P-0006328-T01-IM5 |            | NRBF2-FGFR2 | NRBF2        | in-frame     | five_prime  | 123241245          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |              |                   | FGFR2 (NM_000141) - NRBF2 (NM_030759) rearrangement: c.2302-1710:FGFR2_c.157-364:NRBF2inv   Protein fusion: in frame (FGFR2-NRBF2)                                                            | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains                            | source_annotation_text                                                                                                                    | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>9071-T01-IM6-180 | P-0049<br>071-T01-IM6 |            | NRG3-FGFR2  | NRG3         | out-of-frame | thre_prime  | 123244882          | 17              | 732                         | True                   | disrupted     | 0.09386281588447654      | True                |                                              | Immunoglobulin domain                        | Protein tyrosine and serine/threonine kinase | NRG3 (NM_001010848) - FGFR2 (NM_000141) rearrangement: c.823+108972:NRG3_c.2195+27:FGFR2inv   Protein Fusion: out of frame {NRG3:FGFR2}   | NA                           |
| EVT-P-002<br>2337-T01-IM6-181 | P-0022<br>337-T01-IM6 |            | NRG3-FGFR2  | NRG3         | out-of-frame | thre_prime  | 123241723          | 18              | 768                         | True                   | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase |                                              | NRG3 (NM_001010848) - FGFR2 (NM_000141) Rearrangement : c.953+89877:NRG3_c.2301+1489:FGFR2inv   Protein Fusion: out of frame {NRG3:FGFR2} | NA                           |
| EVT-P-003<br>2171-T03-IM6-182 | P-0032<br>171-T03-IM6 |            | OPTN-FGFR2  | OPTN         | in-frame     | five_prime  | 123239789          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                                              | FGFR2 (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv   Protein Fusion: in frame {FGFR2:OPTN}           | NA                           |

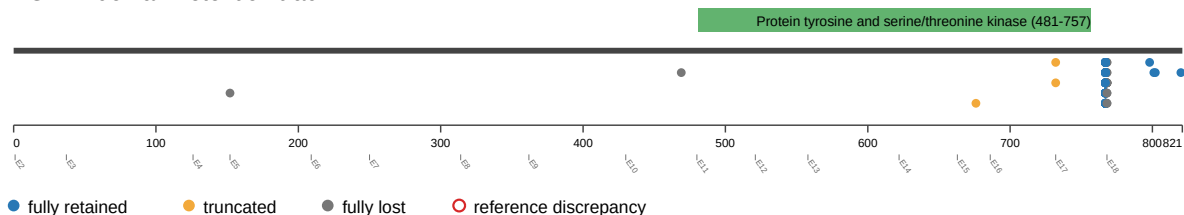
| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name      | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains                                                                                                                                                            | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                             | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|----------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-003<br>2171-T01-I<br>M6-183 | P-0032<br>171-T0<br>1-IM6 |                            | OPTN-<br>FGFR2       | OPT<br>N                 | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123239<br>789              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                                 |                           | FGFR2 (NM_000141) -<br>OPTN (NM_021980)<br>rearrangement: c.2302-254:<br>FGFR2_c.553-1543:OPTNin<br>v   Protein Fusion: in frame<br>{FGFR2:OPTN}                                   | NA                                   |
| EVT-P-002<br>2781-T01-I<br>M6-184 | P-0022<br>781-T0<br>1-IM6 |                            | PDZR<br>N4-FG<br>FR2 | PDZ<br>RN4               | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123241<br>997              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                                 |                           | FGFR2 (NM_000141) -<br>PDZRN4 (NM_001164595)<br>rearrangement: t(6;10)(q23.1<br>;q26.1)(chr6:g.130408449::c<br>hr10:g.123239680)   Protein<br>Fusion: in frame<br>{FGFR2:PDZRN4}   | NA                                   |
| EVT-P-004<br>4768-T01-I<br>M6-187 | P-0044<br>768-T0<br>1-IM6 |                            | RNF14<br>5-FGF<br>R2 | RNF<br>145               | out-fr<br>ame                | thre<br>e_<br>pr<br>ime | 123243<br>127              | 18                          | 768                                 | True                           | lost                      | 0.0                              | False                       |                                                                                                                                                           | Prote<br>in tyr<br>osine<br>and<br>serin<br>e/thr<br>eoni<br>ne ki<br>nase<br>; Im<br>mun<br>oglo<br>bulin<br>dom<br>ain; I<br>mmu<br>nogl<br>obuli<br>n<br>I-set<br>dom<br>ain |                           | RNF145 (NM_144726) -<br>FGFR2 (NM_000141)<br>rearrangement: t(5;10)(q33.3<br>;q26.13)(chr5:g.158598146::<br>chr10:g.123243127)   Protein<br>Fusion: out of frame<br>{RNF145:FGFR2} | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_name     | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_r<br>ole | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intro<br>nic_bre<br>akpoint | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains                                                                                                                                  | lost_<br>dom<br>ains                                                                                                                                                    | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                                        | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|---------------------|--------------------------|------------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-000<br>6235-T01-I<br>M5-188 | P-0006<br>235-T0<br>1-IM5 |                            | ROCK<br>1-FGF<br>R2 | RO<br>CK1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>133              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                         |                           | FGFR2 (NM_000141) -<br>ROCK1 (NM_005406)<br>rearrangement: t(10;18)(q26.<br>13;q11.1)(chr10:g.12324013<br>3::chr18:g18680051)  <br>Protein fusion: in frame<br>(FGFR2-ROCK1)                  | NA                                   |
| EVT-P-003<br>9370-T01-I<br>M6-189 | P-0039<br>370-T0<br>1-IM6 |                            | RPAP3<br>-FGFR<br>2 | RPA<br>P3                | in-fr<br>ame                 | thr<br>ee_<br>pri<br>me | 123242<br>832              | 18                          | 768                                 | True                           | lost                      | 0.0                              | False                       |                                                                                                                                                           | Prote<br>in tyr<br>osine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Im<br>mun<br>oglo<br>bulin<br>dom<br>ain; I<br>mmu<br>nogl<br>obuli<br>n<br>I-set<br>dom<br>ain |                           | RPAP3 (NM_024604) -<br>FGFR2 (NM_000141)<br>rearrangement: t(10;12)(q26.<br>13;q13.11)(chr10:g.1232428<br>32::chr12:g.48092574)  <br>Protein Fusion: in frame<br>{RPAP3:FGFR2}                | NA                                   |
| EVT-P-002<br>5074-T01-I<br>M6-190 | P-0025<br>074-T0<br>1-IM6 |                            | SHTN1<br>-FGFR<br>2 | SHT<br>N1                | in-fr<br>ame                 | fiv<br>e_<br>pri<br>me  | 123240<br>122              | 17                          | 767                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protei<br>n tyros<br>ine<br>and se<br>rine/thr<br>eonine<br>kinase<br>; Immu<br>noglob<br>ulin do<br>main; I<br>mmun<br>oglobu<br>lin<br>I-set d<br>omain |                                                                                                                                                                         |                           | FGFR2 (NM_000141) -<br>KIAA1598 (NM_001127211)<br>Fusion (FGFR2 exon 17 with<br>KIAA1598 exon 5): c.2302-5<br>87:FGFR2_c.267+2596:KIA<br>A1598   Protein Fusion: in<br>frame {FGFR2:KIAA1598} | NA                                   |

| event_id                          | sample_id                 | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                                                  | lost_domains                                                                                      | disrupted_domains | source_annotation_text                                                                                                                                       | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>2507-T01-I<br>M6-191 | P-0032<br>507-T0<br>1-IM6 |            | SORBS1-FGFR2 | SORBS1       | unknown      | five_prime  | 123242268          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |                                                                                                   |                   | FGFR2 (NM_000141) Rearrangement : c.2301+944:FGFR2_chr10:g.97250518del   Transcript Fusion {FGFR2:SORBS1}                                                    | NA                            |
| EVT-P-002<br>2337-T01-I<br>M6-192 | P-0022<br>337-T0<br>1-IM6 |            | TBC1D4-FGFR2 | TBC1D4       | in-frame     | five_prime  | 123241723          | 17              | 767                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |                                                                                                   |                   | FGFR2 (NM_000141) - TBC1D4 (NM_014832) Rearrangement : t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946348)   Protein Fusion: in frame {FGFR2:TBC1D4} | NA                            |
| EVT-P-002<br>1788-T01-I<br>M6-193 | P-0021<br>788-T0<br>1-IM6 |            | VPS26A-FGFR2 | VPS26A       | in-frame     | three_prime | 123242226          | 18              | 768                         | True                   | lost          | 0.0                      | False               |                                                                                                   | Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain |                   | VPS26A (NM_004896) - FGFR2 (NM_000141) Rearrangement : c.4-1035:VPS26A_c.2301+986:FGFR2inv   Protein Fusion: in frame {VPS26A:FGFR2}                         | NA                            |

## domain retention outliers

FGFR2 domain-retention track



## fusion schematic

FGFR2 fusion-transcript schematic

partner block → breakpoint → retained FGFR2 portion (domain-colored, exon ticks)



Showing the top 28 of 84 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk\_impact\_50k\_2026

